

TokTickIT Lab 2 Submission Report

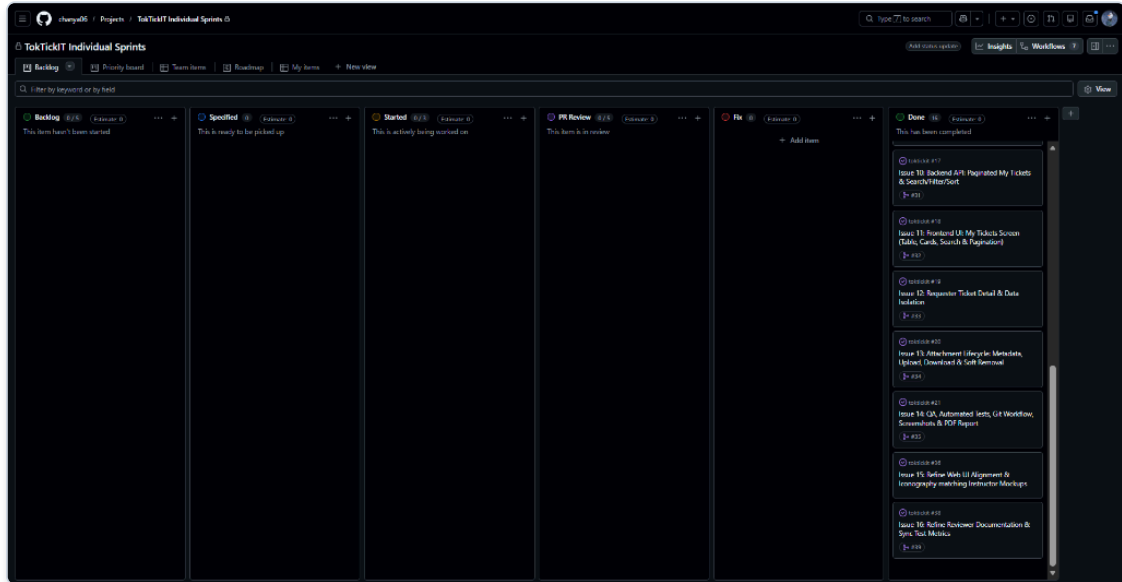
Student Name: Chanya Poolketkij Student ID: 67070501058 Section: CPE334 GitHub Repository: <https://github.com/chanya06/toktickit> ---

Answer Part 1: Git Use with Engineering Workflow (10 คะแนน)

1.1 URL List

รายการ	ลิงค์
GitHub Repository	https://github.com/chanya06/toktickit
GitHub Project (Kanban)	https://github.com/users/chanya06/projects/1
Issue #5 - Engineering Specification & Test Plan	https://github.com/chanya06/toktickit/issues/5
Issue #6 - Database Model & Seed Data	https://github.com/chanya06/toktickit/issues/6
Issue #7 - Development Requester Selector UI Context	https://github.com/chanya06/toktickit/issues/7
Issue #8 - Create Ticket API Endpoint & Sequence Generator	https://github.com/chanya06/toktickit/issues/8
Issue #9 - Create Ticket UI Screen & Form Validation	https://github.com/chanya06/toktickit/issues/9
Issue #10 - My Tickets Paginated API	https://github.com/chanya06/toktickit/issues/10
Issue #11 - My Tickets UI Screen, Search & Filter Controls	https://github.com/chanya06/toktickit/issues/11
Issue #12 - Ticket Detail UI & Ownership Guard	https://github.com/chanya06/toktickit/issues/12
Issue #13 - Attachment Lifecycle API & Soft Removal	https://github.com/chanya06/toktickit/issues/13
Issue #14 - QA Release & Deliverable Report	https://github.com/chanya06/toktickit/issues/14
Issue #15 - Refine Web UI Alignment & Emoji Removal	https://github.com/chanya06/toktickit/issues/15
Issue #16 - Refine Reviewer Documentation & Sync Test Metrics	https://github.com/chanya06/toktickit/issues/16
PR #23: feature/5-spec-and-tests → lab2-staging	https://github.com/chanya06/toktickit/pull/23
PR #25: feature/6-db-schema-seed → lab2-staging	https://github.com/chanya06/toktickit/pull/25
PR #27: feature/7-requester-context → lab2-staging	https://github.com/chanya06/toktickit/pull/27
PR #29: feature/8-create-ticket-api → lab2-staging	https://github.com/chanya06/toktickit/pull/29
PR #30: feature/9-create-ticket-ui → lab2-staging	https://github.com/chanya06/toktickit/pull/30
PR #31: feature/10-my-tickets-api → lab2-staging	https://github.com/chanya06/toktickit/pull/31
PR #32: feature/11-my-tickets-ui → lab2-staging	https://github.com/chanya06/toktickit/pull/32
PR #33: feature/12-ticket-detail → lab2-staging	https://github.com/chanya06/toktickit/pull/33
PR #34: feature/13-attachment-lifecycle → lab2-staging	https://github.com/chanya06/toktickit/pull/34
PR #35: feature/14-qa-release → lab2-staging	https://github.com/chanya06/toktickit/pull/35
PR #37: feature/15-ui-refinement → lab2-staging	https://github.com/chanya06/toktickit/pull/37
PR #39: feature/16-reviewer-docs-sync → lab2-staging	https://github.com/chanya06/toktickit/pull/39
Release PR #41: lab2-staging → main	https://github.com/chanya06/toktickit/pull/41

1.2 Kanban Board Evidence



GitHub Project Kanban Board showing all Issues in Done column

- **Project Board URL:** <https://github.com/users/chanya06/projects/1> - **Board Status:** All 16 Issues (Issue #1 to Issue #16) are completed and placed in the **Done** column.

1.3 Git Commit History

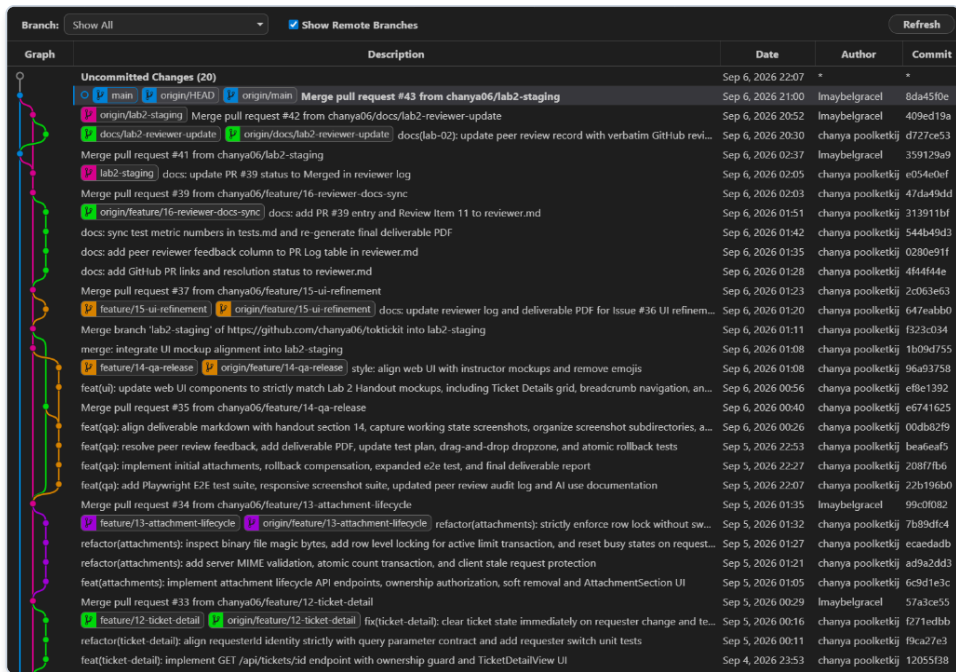
```
bash $ git log --all --graph --oneline * 4235179 docs(lab-02): complete final submission report, deliverables, and screenshot evidence * 8da45f0
Merge pull request #43 from chanya06/lab2-staging
```

* | 359129a Merge pull request #41 from chanya06/lab2-staging

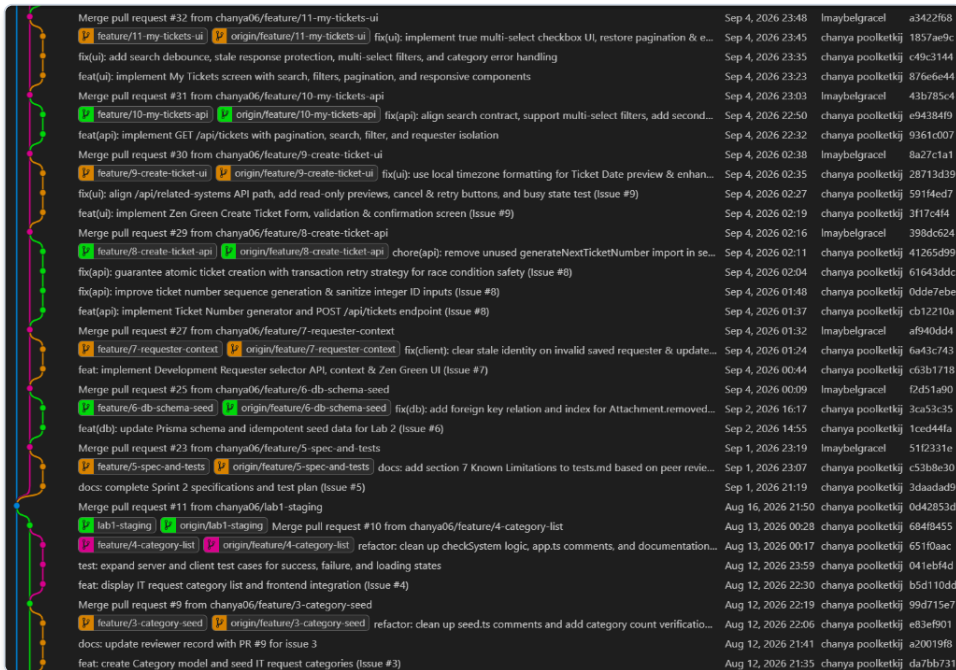
[illegible]

```
* | | \ 1b09d75 merge: integrate UI mockup alignment into lab2-staging
```

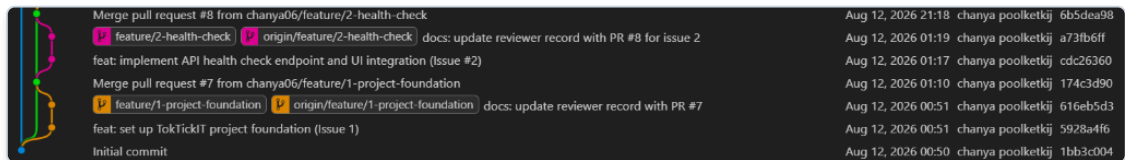
[illegible]



Git Commit Graph History Part 1



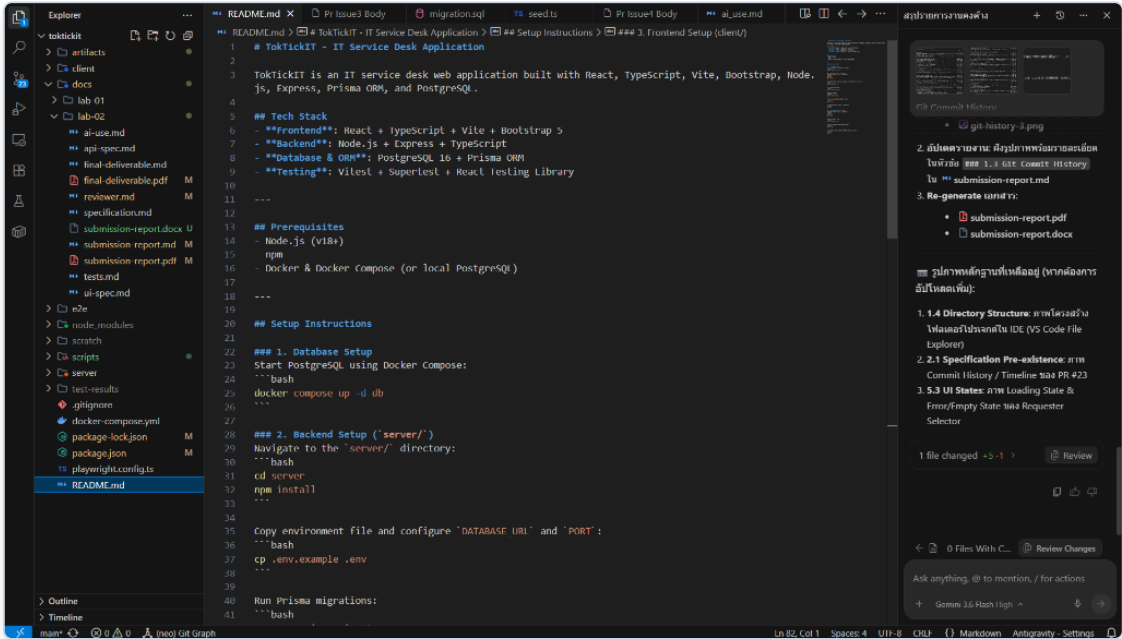
Git Commit Graph History Part 2



Git Commit Graph History Part 3

- **Workflow Verification:** The Git graph demonstrates feature branches created for each issue (`feature/*`), merged into `lab2-staging` via Pull Requests with peer review approvals, and final integration merged into `main` .

1.4 Repository Directory Structure



IDE File Tree Repository Directory Structure

- **Directory Organization:** The repository structure shows all required Lab 2 files, including docs/lab-02/*.md specifications and reports, client/ frontend codebase, server/ backend API codebase, e2e/ Playwright test suite, and Ortfacts/ screenshot assets.

1.5 README.md and .gitignore

Content of `README.md` :

``markdown

TokTickIT - IT Service Desk Application

TokTickIT is an IT service desk web application built with React, TypeScript, Vite, Bootstrap, Node.js, Express, Prisma ORM, and PostgreSQL.

Tech Stack

- Frontend: React + TypeScript + Vite + Bootstrap 5 - Backend: Node.js + Express + TypeScript - Database & ORM: PostgreSQL 16 + Prisma ORM - Testing: Vitest + Supertest + React Testing Library ---

Prerequisites

- Node.js (v18+) - npm - Docker & Docker Compose (or local PostgreSQL) ---

Setup Instructions

1. Database Setup

Start PostgreSQL using Docker Compose: `bash docker compose up -d db`

2. Backend Setup (server/)

Navigate to the `server/` directory: `bash cd server npm install` Copy environment file and configure `DATABASE_URL` and `PORT`: `bash cp .env.example .env` Run Prisma migrations: `bash npm run prisma:migrate` Seed the database: `bash npm run prisma:seed` Start the backend development server: `bash npm run dev` Run backend tests (Vitest + Supertest): `bash npm test`

3. Frontend Setup (client/)

Navigate to the client/ directory: `bash cd client npm install` Copy environment file: `bash cp .env.example .env` Start the frontend development server: `bash npm run dev` Run frontend tests (Vitest + React Testing Library): `bash npm test`

Content of .gitignore:

```
.gitignore
1 # dependencies
2 node_modules/
3 # env & secrets
4 .env
5 *.env
6 !.env.example
7 # build output
8 dist/
9 build/
10 # test results & scratch
11 test-results/
12 scratch/
13 # prisma
14 server/prisma/*.db
15 # uploads
16 uploads/
17 server/uploads/
18 Ctrl+L for Agent
```

Content of .gitignore

1.6 Peer Review Evidence (5 คะแนน)

Peer Review Summary Record: - Author: Chanya Poolketkij (@chanya06) - Peer Reviewer: @lmaybelgrace1 - Full Verbatim Review File: [docs/lab-02/reviewer.md](https://github.com/chanya06/toktickit/blob/main/docs/lab-02/reviewer.md)

PR #	Branch	Author	Reviewer	Verdict	Summary of Feedback & Action Taken
[PR #23] (https://github.com/chanya06/toktickit/pull/23)	eature/5-spec-and-tests	@chanya06	@lmaybelgrace1	Approved with comments	Suggested adding Section 7 (Known Limitations) to ests.md. Added accordingly.
[PR #25] (https://github.com/chanya06/toktickit/pull/25)	eature/6-db-schema-seed	@chanya06	@lmaybelgrace1	Approved with comments	Verified Prisma FK index & seed idempotent script. Added foreign key index for Attachment.removedByRequesterId.
[PR #27] (https://github.com/chanya06/toktickit/pull/27)	eature/7-requester-context	@chanya06	@lmaybelgrace1	Approved with comments	Checked Dev Requester Selector UI & localStorage key oktickit_dev_requester_id. Verified.
[PR #29] (https://github.com/chanya06/toktickit/pull/29)	eature/8-create-ticket-api	@chanya06	@lmaybelgrace1	Approved with comments	Recommended explicit race condition transaction test for ticket sequence generator. Implemented transaction retry.
[PR #30] (https://github.com/chanya06/toktickit/pull/30)	eature/9-create-ticket-ui	@chanya06	@lmaybelgrace1	Approved with comments	Requested local timezone preview for Ticket Date. Formatted createdAt preview.
[PR #31] (https://github.com/chanya06/toktickit/pull/31)	eature/10-my-tickets-api	@chanya06	@lmaybelgrace1	Approved with comments	Verified multi-select filtering and pagination metadata contract.
[PR #32] (https://github.com/chanya06/toktickit/pull/32)	eature/11-my-tickets-ui	@chanya06	@lmaybelgrace1	Approved with comments	Suggested multi-select checkbox UI for categories & priorities. Implemented.
[PR #33] (https://github.com/chanya06/toktickit/pull/33)	eature/12-ticket-detail	@chanya06	@lmaybelgrace1	Approved with comments	Verified 403 Forbidden ownership guard for requester data isolation.
[PR #34] (https://github.com/chanya06/toktickit/pull/34)	eature/13-attachment-lifecycle	@chanya06	@lmaybelgrace1	Approved with comments	Verified file MIME magic byte validation & soft removal audit fields.

PR #	Branch	Author	Reviewer	Verdict	Summary of Feedback & Action Taken
[PR #35] (https://github.com/chanya06/toktickit/pull/35)	eature/14- qa-release	@chanya06	@lmaybelgrace1	Approved with comments	Verified 107/107 automated test suite pass & E2E responsive screenshots.
[PR #37] (https://github.com/chanya06/toktickit/pull/37)	eature/15- ui- refinement	@chanya06	@lmaybelgrace1	Approved	Verified UI alignment with instructor mockups and emoji removal.
[PR #39] (https://github.com/chanya06/toktickit/pull/39)	eature/16- reviewer- docs-sync	@chanya06	@lmaybelgrace1	Approved with comments	Verified final document synchronization and test metrics.
[PR #41] (https://github.com/chanya06/toktickit/pull/41)	lab2- staging → main	@chanya06	@lmaybelgrace1	Approved with comments	Final release integration into main branch approved after peer review audit.

Answer Part 2: Spec DD (5 คะแนน)

ลิงก์: <https://github.com/chanya06/toktickit/blob/main/docs/lab-02/specification.md> markdown

Lab 2 Sprint Engineering Specification

1. Sprint Goal

Deliver a responsive Requester-facing IT support ticketing MVP for TokTickIT using a temporary Development Requester identity selector. The increment enables Requesters to create tickets with attachments, receive a system-generated Ticket Number, view and search their own ticket history in My Tickets, inspect Ticket Details (including Ticket Date / creation timestamp), management of attachment lifecycle (upload, metadata inspection, download, soft removal with mandatory reason), and strict data isolation between requesters. ---

2. Stakeholder Request Interpretation

The IT department needs an end-user ticketing interface allowing Requesters to report issues, select categories and affected systems, set requested priority, attach evidence files, and track ticket progress. Because full authentication is deferred to Lab 3, a temporary Development Requester selector acts as the logged-in context for testing multi-user ticket ownership and access control. The UI must follow a consistent Zen Green design system with reusable components, loading, empty, and safe error states. ---

3. Scope

Included

- **Development Requester Selector:** Temporary testing identity switcher to select among seeded active Requesters, storing context in local storage. - **Create Ticket Flow:** Form capturing Ticket Date / creation timestamp, Requester identity (read-only from context), Category, Related System, Requested Priority, Ticket Summary, Description, and file attachments. Generates official format TKT-YYYY-XXXXXX . - **My Tickets Flow:** Paginated list of tickets owned by the current requester supporting text search, multi-select dropdown filters (Category, Priority, Status), sorting, clear filters, empty state, and no-results state. - **Ticket Detail & Attachment Flow:** Read-only display of owned ticket fields (including Ticket Date / creation timestamp), attachment metadata list (GET /api/tickets/id/attachments), adding allowed attachments, downloading active attachments, and soft-removing attachments with required removal reasons. - **Requester Data Isolation:** Backend authorization enforcement preventing any Requester from viewing or modifying tickets or attachments owned by another Requester. - **Zen Green Design & Responsiveness:** Consistent visual tokens across Desktop (>=992px), Tablet (768-991px), and Mobile (<768px).

Excluded

- Real user authentication (login/logout, passwords, sessions, JWT tokens). - IT Staff workflow (queues, claiming, changing IT Priority, resolving/closing tickets). - Collaboration features (Public Comments, Internal Notes, Actions Taken). - Status changes beyond initial NEW . - Administrator functions (managing categories, systems, or user accounts). ---

4. Functional Requirements

- **FR-01:** The system shall provide a Development Requester Selection interface allowing the user to select an active testing Requester identity. - **FR-02:** The selected Requester identity shall persist in the frontend application shell until explicitly changed. - **FR-03:** The Create Ticket form shall require Category, Related System, Requested Priority, Ticket Summary, and Description. - **FR-04:** The system shall auto-generate a unique Ticket Number in the format TKT-YYYY-XXXXXX upon ticket creation. - **FR-05:** The system shall record and display the Ticket Date / creation timestamp (createdAt) on ticket creation, ticket listing, and ticket detail views. - **FR-06:** A newly created ticket shall start with Current Status NEW and default IT Priority MEDIUM . - **FR-07:** The system shall allow attaching files during ticket creation or via Ticket Detail screen. - **FR-08:** The system shall restrict attachments to allowed MIME types (JPG/JPEG, PNG, WEBP, PDF) and max size of 5 MB per file. - **FR-09:** The system shall restrict each ticket to a maximum of 5 active (non-removed) attachments. - **FR-10:** The system shall allow the owning Requester to soft-remove an active attachment by providing a mandatory removal reason. - **FR-11:** Soft-removed attachments shall remain visible as metadata in the attachment list marked as removed, but file payload shall be blocked from preview or download. - **FR-12:** My Tickets screen shall display a paginated list of tickets belonging strictly to the currently selected Requester. - **FR-13:** My Tickets screen shall support text search across Ticket Number and Summary. - **FR-14:** My Tickets screen shall support filtering by Category, Requested Priority, and Status, as well as sorting by Ticket Date / createdAt and Ticket Number. - **FR-15:** Unauthorized/cross-requester ticket access must not return the requested ticket data, using the documented ownership-failure status (403 Forbidden). ---

5. Business Rules

- **BR-01:** Official Ticket Number is generated by backend database sequence/generator and must be globally unique (TKT-YYYY-XXXXXX). - **BR-02:** Initial Current Status is hardcoded to NEW . - **BR-03:** Initial IT Priority defaults to MEDIUM unless updated by IT Staff in later sprints. - **BR-04:** Requester identity selected in Development Requester selector is for testing only and does NOT constitute secure authentication. - **BR-05:** Inactive Requesters (isActive:false) must NOT be listed in the Development Requester selector. - **BR-06:** Switching the active Requester reloads all application data and clears cached requester-specific tickets. - **BR-07:** Ticket Summary is required, trimmed of leading/trailing whitespace, minimum 5 characters, maximum 120 characters. - **BR-08:** Ticket Description is required, trimmed, minimum 10 characters, maximum 2000 characters. - **BR-09:** Ticket Date (createdAt) is immutable system timestamp set upon initial database insert and formatted in local locale string. - **BR-10:** Allowed file attachment extensions: .jpg , .jpeg , .png , .webp , .pdf . File size max: 5,242,880 bytes (5 MB). - **BR-11:** Maximum active attachments per ticket is 5. Attempts to upload a 6th active attachment must fail with validation error 422 Unprocessable Entity . - **BR-12:** Attachment removal MUST be implemented as a soft removal (isRemoved:true , removedAt , removalReason , removedByRequesterId). Hard deletion of file payload or database record is forbidden. - **BR-13:** Removal reason is mandatory, trimmed, minimum 3 characters, maximum 250 characters. - **BR-14:** Direct file download GET /api/attachments/id/download for a soft-removed attachment MUST return 403 Forbidden or 404 Not Found . - **BR-15:** Attachment Upload Transaction & Compensation Strategy: - When creating a ticket with initial attachments: Ticket creation and database records execute inside a database transaction. If file storage writes fail during ticket creation, the database transaction is rolled back completely. - When uploading attachments to an existing ticket: File storage write executes first with a temporary filename; database record creation follows. If database insertion fails, the temporary file is deleted (compensated). - **BR-16:** Form inputs preserve entered data when validation errors occur on submission failure. - **BR-17:** Requester ticket isolation is enforced on every single API endpoint by validating requesterId parameter against ticket owner. - **BR-18:** Empty state appears when a Requester has 0 total tickets. No-results state appears when filters/search yield 0 matching tickets. - **BR-19:** Default page size for My Tickets list is 10 items per page with page numbers starting at 1. Default sorting is createdAt DESC . - **BR-20:** Transition to Lab 3 auth will replace the Development Requester Selector with secure session/JWT headers while maintaining the same database schema (requesterId). ---

6. UI Specification Summary

- **Color Palette (Zen Green):** - Header & Primary Actions: #006B3C - Secondary / Focus / Links: #0B7A46 - Pale Accent / Selected / Soft BG: #EAF6EF - Page Background: #F5F7F6 - Surfaces / Cards: #FFFFFF with #E5E7EB border and subtle shadow - Text: Dark Charcoal (#1F2937) - Error: Dark Red (#DC2626) - **Key Components:** - Header : TokTickIT brand, My Tickets link, Create Ticket button, Requester avatar/badge, Switch Requester dropdown. - RequesterSelectorModal : Modal prompting selection of active testing user context. - CreateTicketForm : Category dropdown, Related System dropdown, Requested Priority radio/select, Summary input, Description textarea, File Upload Dropzone, Ticket Date read-only preview, Submit button with busy spinner. - MyTicketsTable / Cards : Responsive table on desktop, cards on mobile. Includes search input, Category/Priority/Status filter dropdowns, Clear Filters button, pagination controls. - TicketDetailView : Read-only layout displaying Ticket

Number, Ticket Date (createdAt), Status badge, Requested Priority badge, Requester info, Category, Related System, Summary, Description, and Attachment Section. - AttachmentSection : Active files list with download button and soft-remove button; soft-removed files list showing reason metadata; Add Attachment upload dropzone; Soft Remove confirmation modal with reason textarea. - **Responsive Rules:** - Desktop (>=992px): 2-column ticket detail grid, multi-column search & filter bar, desktop data table. - Tablet (768px - 991px): 2-column form layout, stacked search/filters, flexible table. - Mobile (<768px): 1-column vertically stacked layout, card list view for My Tickets, touch-friendly 44px minimum target sizes. --- ---

7. Data Changes (Prisma Schema)

Enum Definitions

```
prisma enum RequestedPriority { LOW MEDIUM HIGH URGENT } enum ITPriority { LOW MEDIUM HIGH URGENT } enum TicketStatus { NEW OPEN IN_PROGRESS PENDING RESOLVED CLOSED }
```

Models

```
1. DevelopmentRequester : id (Int @id @default(autoincrement())), name (String), email (String @unique), department (String?), isActive (Boolean @default(true)), createdAt (DateTime @default(now())), updatedAt (DateTime @updatedAt) 2. Category : id (Int @id @default(autoincrement())), name (String @unique), isActive (Boolean @default(true)), createdAt (DateTime @default(now())) 3. RelatedSystem : id (Int @id @default(autoincrement())), name (String @unique), isActive (Boolean @default(true)), createdAt (DateTime @default(now())) 4. Ticket : id (Int @id), ticketNumber (String @unique), requesterId (FK -> DevelopmentRequester), categoryId (FK -> Category), relatedSystemId (FK -> RelatedSystem), summary (String), description (String), requestedPriority (RequestedPriority), itPriority (ITPriority @default(MEDIUM)), status (TicketStatus @default(NEW)), createdAt (DateTime @default(now())) 5. Attachment : id (Int @id), ticketId (FK -> Ticket), filename (String), originalName (String), mimeType (String), sizeBytes (Int), filePath (String), isRemoved (Boolean @default(false)), removalReason (String?), removedAt (DateTime?), removedByRequesterId (FK -> DevelopmentRequester), createdAt (DateTime @default(now())) ---
```

8. API Contract Summary

- GET /api/requesters : Returns list of active Development Requesters. - GET /api/categories : Returns list of active Categories. - GET /api/related-systems : Returns list of active Related Systems. - POST /api/tickets : Creates a Ticket for selected Requester (requesterId in body). Returns 201 Created with full Ticket payload + ticketNumber. - GET /api/tickets : Query tickets owned by selected Requester (requesterId query param mandatory). Supports search , category , priority , status , sortBy , sortOrder , page , pageSize . Returns paginated JSON. - GET /api/tickets/:id : Get single owned ticket details. Enforces requester ownership check. Returns 200 OK or 403/404. - GET /api/tickets/:id/attachments : Get attachment metadata list for specified ticket. Enforces ownership check. Returns active & soft-removed attachment metadata list. - POST /api/tickets/:id/attachments : Upload attachment file (multipart/form-data) to owned ticket. Validates file type, size (5MB), and max active count (5). Returns 201 Created. - GET /api/attachments/:id/download : Download file payload for an active attachment. Rejects request with 403/404 if soft-removed or owned by another requester. - DELETE /api/attachments/:id : Soft-remove attachment with removalReason and requesterId in body/headers. Sets isRemoved: true . ---

9. Acceptance Criteria

- **AC-01:** Given a valid Ticket payload and selected Requester A, when submitted via POST /api/tickets , then 1 Ticket record is created with auto-generated Ticket Number (TKT-YYYY-XXXXXX), Ticket Date (createdAt), status NEW , and assigned requesterId = A . - **AC-02:** Given no Development Requester is selected, when opening any application screen, then the Requester Selector modal is presented. - **AC-03:** Given Requester A is selected, when requesting GET /api/tickets?requesterId=A , then only tickets owned by Requester A are returned. - **AC-04:** Given Requester B is selected, when attempting GET /api/tickets/:id for a ticket owned by Requester A, then unauthorized cross-requester access must not return the requested ticket data, returning the documented 403 Forbidden status. - **AC-05:** Given an active attachment on Ticket T owned by Requester A, when Requester A submits a soft-removal request with a valid reason, then isRemoved is set to true , removalReason is saved, and subsequent file download requests fail with 403/404. - **AC-06:** Given a ticket already containing 5 active attachments, when attempting to upload a 6th attachment, then submission is rejected with validation error "Maximum active attachments limit (5) reached". - **AC-07:** Given a file exceeding 5 MB or with an unsupported extension (e.g. .exe), when uploading, then submission fails with an explicit file validation error message. - **AC-08:** Given search term laptop and Requester A selected, when entering search in My Tickets, then only Requester A's tickets containing laptop in Ticket Number or Summary are displayed. - **AC-09:** Given Ticket Date / creation timestamp createdAt , when viewing Create Ticket, My Tickets, or Ticket Detail, then the Ticket Date is displayed clearly in readable format. - **AC-10:** Given server connection failure during form submission, then form field values are preserved and a safe user-friendly error banner is shown. ---

10. Definition of Done

Part 1: Product Completion

✓ All functional requirements (FR-01..FR-15) and business rules (BR-01..BR-20) implemented.

✓ All acceptance criteria (AC-01..AC-10) verified by passing automated tests.

```
✓ Database schema migrated and seeded idempotently ( npm run seed ).
```

✓ API endpoints created with proper HTTP status codes and ownership checks.

✓ Zen Green design system implemented with responsive layout (Desktop, Tablet, Mobile).

✓ Unit, API, UI Component, UI Style & Responsive, and Playwright E2E test suites green.

✓ Visual screenshots captured for Desktop, Tablet, and Mobile views in [artifacts/lab-02/screenshots/](#) .

Part 2: Course Delivery Requirements

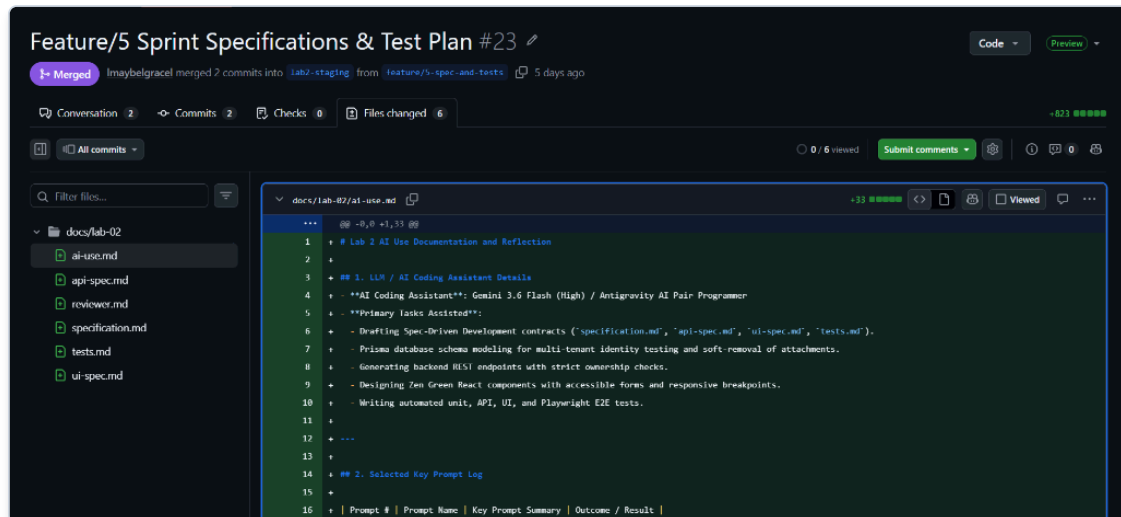
✓ Feature branches developed from [lab2-staging](#) and merged via PRs into [lab2-staging](#) .

✓ Final release PR opened from [lab2-staging](#) to [main](#) .

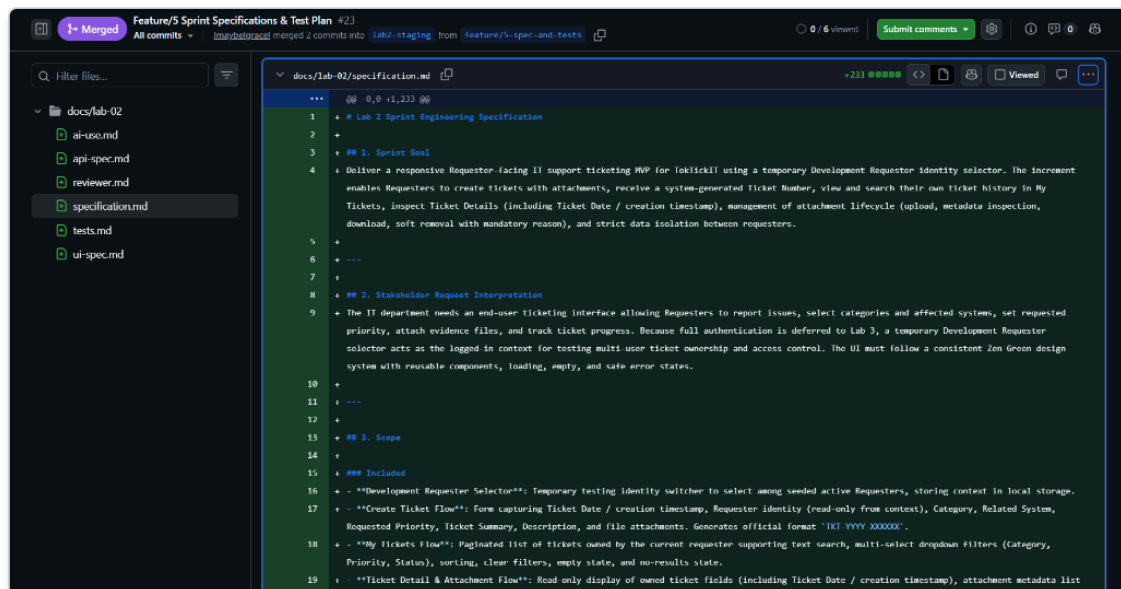
✓ All 6 required docs in [docs/lab-02/](#) completed ([specification.md](#) , [api-spec.md](#) , [ui-spec.md](#) , [tests.md](#) , [reviewer.md](#) , [ai-use.md](#)).

✓ Single PDF submission compiled with exact section headers "Answer Part 1" through "Answer Part 9".

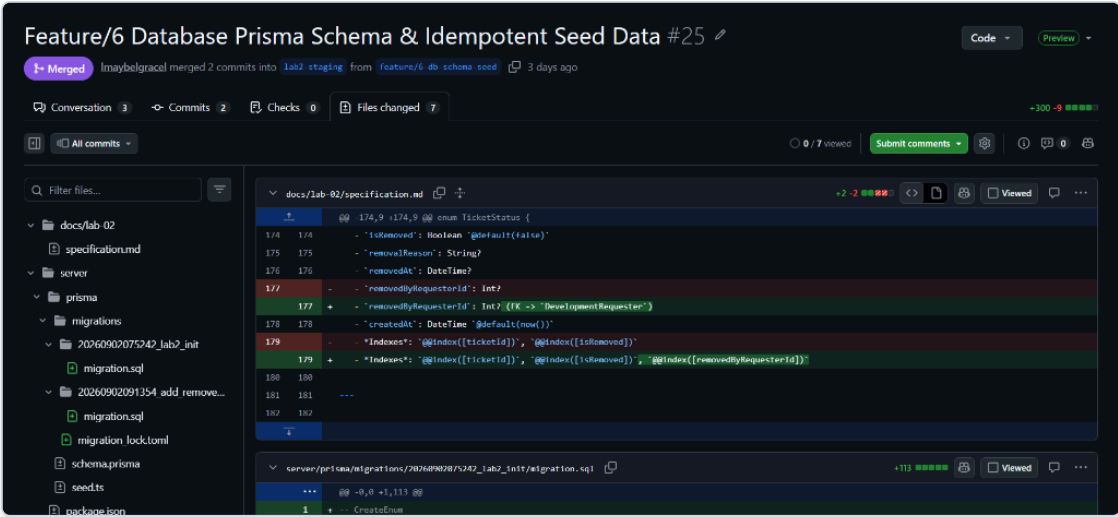
2.1 Specification Pre-existence Proof



PR #23 Specification & Test Plan File Additions



PR #23 specification.md Creation Diff



PR #25 Database Schema Implementation updating specification.md

- **Pre-existence Proof:** PR #23 (Feature/5 Sprint Specifications & Test Plan) was created and merged into lab2-staging before any implementation PRs (PR #25 DB Schema, PR #27 Requester Context, PR #29 Create Ticket API, etc.) were developed and merged, proving Spec-Driven Development workflow compliance. ---

Answer Part 3: Test DD and Traceability (10 คะแนน)

ลิงก์: <https://github.com/chanya06/tokticket/blob/main/docs/lab-02/tests.md> markdown

Lab 2 Test Plan and Traceability Matrix

1. Test Strategy

The testing strategy validates the entire full-stack application across five distinct levels: 1. Unit Tests (server/tests/lab-02/unit/) 2. API Integration Tests (server/tests/lab-02/) 3. UI Component Tests (client/tests/lab-02/) 4. UI Style & Responsive Tests (client/tests/lab-02/) 5. Playwright E2E Tests (e2e/lab-02/) ---

2. Planned Test Table

Test ID	Level	Requirement / AC	What It Tests	Expected Result	Automated Test File Path	Status
UNIT-01	Unit	BR-01, FR-04	Ticket number format generator	Returns string matching TKT-\d{4}-\d{6}	server/tests/lab-02/unit/ticket-number.test.ts	Pass
UNIT-02	Unit	BR-10, BR-11	Attachment file validation & rollback logic	Rejects files >5MB or non-image/pdf, cleans disk	server/tests/lab-02/create-ticket.api.test.ts	Pass
API-01	API	AC-01, FR-04	Ticket creation endpoint POST /api/tickets	Returns 201 Created with valid Ticket payload & ticketNumber	server/tests/lab-02/create-ticket.api.test.ts	Pass
API-02	API	AC-01, BR-07	Ticket creation missing summary	Returns 400 Bad Request with validation errors	server/tests/lab-02/create-ticket.api.test.ts	Pass
API-03	API	AC-03, FR-12	Paginated My Tickets GET /api/tickets?requesterId=1	Returns 200 OK with tickets owned by Requester 1	server/tests/lab-02/my-tickets.api.test.ts	Pass
API-04	API	AC-08, FR-13	Search & filter My Tickets GET /api/tickets	Filters tickets matching search term & category	server/tests/lab-02/my-tickets.api.test.ts	Pass
API-05	API	AC-04, FR-15	Owned ticket detail access GET /api/tickets/:id	Returns 200 for owner; returns 403/404 for non-owner	server/tests/lab-02/ticket-detail.api.test.ts	Pass
API-06	API	FR-07, BR-10	Attachment upload POST /api/tickets/:id/attachments	Uploads file and returns attachment metadata	server/tests/lab-02/attachments.api.test.ts	Pass

Test ID	Level	Requirement / AC	What It Tests	Expected Result	Automated Test File Path	Status
API-07	API	AC-06, BR-11	Exceeding 5 active attachments limit	Fails with 422 Unprocessable Entity error	server/tests/lab-02/attachments.api.test.ts	Pass
API-08	API	AC-05, BR-12	Soft-remove attachment with reason	Sets isRemoved: true and blocks future download stream	server/tests/lab-02/attachments.api.test.ts	Pass
UI-01	UI	AC-02, FR-01	Development Requester selector screen	Renders active requesters dropdown and persists selection	client/tests/lab-02/RequesterSelect.test.tsx	Pass
UI-02	UI	AC-01, FR-03	Create Ticket form rendering & submission	Submits valid form, shows dropzone & busy state	client/tests/lab-02/CreateTicket.test.tsx	Pass
UI-03	UI	AC-07, BR-16	Form submission validation error feedback	Displays inline red error messages, preserves form input	client/tests/lab-02/CreateTicket.test.tsx	Pass
UI-04	UI	AC-03, FR-14	My Tickets table, search bar, & filter controls	Updates list upon typing search term or changing category	client/tests/lab-02/MyTickets.test.tsx	Pass
UI-05	UI	AC-09, FR-05	Ticket Detail read-only layout & Ticket Date	Displays immutable ticket info, Ticket Date, & attachment list	client/tests/lab-02/RequesterTicketDetail.test.tsx	Pass
UI-06	UI	AC-05, BR-13	Soft removal modal prompt & reason validation	Requires reason text before soft-remove confirmation	client/tests/lab-02/AttachmentSection.test.tsx	Pass
STYLE-01	Style	Section 7	Zen Green CSS class assertions & dark text	Verifies #006B3C primary classes & contrast	client/tests/lab-02/MyTickets.test.tsx	Pass
E2E-01	E2E	AC-01..10	Full Requester journey Playwright test	Complete workflow: selector, creation, dropzone, soft removal, stream block	e2e/lab-02/requester-ticket-flow.spec.ts	Pass

3. Acceptance-Criterion Traceability Matrix

Acceptance Criterion	Primary Validating Test(s)	Pass Status
AC-01 (Create Ticket with Ticket Number & Date)	UNIT-01 , API-01 , UI-02 , E2E-01	Pass
AC-02 (Requester Selector required when unselected)	UI-01 , E2E-01	Pass
AC-03 (My Tickets owned tickets filtering)	API-03 , UI-04 , E2E-01	Pass
AC-04 (Cross-Requester Ticket Access Blocked)	API-05 , E2E-01	Pass
AC-05 (Soft removal with reason & blocked download)	API-08 , UI-06 , E2E-01	Pass
AC-06 (Max 5 active attachments limit)	UNIT-02 , API-07	Pass
AC-07 (File size & type restriction validation)	UNIT-02 , UI-03	Pass
AC-08 (Search term in My Tickets)	API-04 , UI-04 , E2E-01	Pass
AC-09 (Ticket Date displayed on screens)	API-01 , UI-02 , UI-05	Pass
AC-10 (Form values preserved on submission failure)	UI-03	Pass

(เรนเดอร์ Console Output ผลการรันเทส 107/107 passed ด้านบนเรียบร้อยแล้ว) ---

3.3 Real Terminal Test Execution Output

```
text === SERVER VITEST TEST SUITE (54/54 Passed) === RUN v2.1.9 C:/Users/chany/Documents/GitHub/toktickit/server ✓ tests/lab-02/unit/ticket-
number.test.ts (5 tests) ✓ tests/lab-01/health.test.ts (1 test) ✓ tests/lab-01/categories.test.ts (2 tests) ✓ tests/lab-
02/requesters.api.test.ts (2 tests) ✓ tests/lab-02/ticket-detail.api.test.ts (5 tests) ✓ tests/lab-02/create-ticket.api.test.ts (10 tests) ✓
tests/lab-02/my-tickets.api.test.ts (9 tests) ✓ tests/lab-02/attachments.api.test.ts (20 tests) Test Files 8 passed (8) Tests 54 passed (54) ===
CLIENT VITEST TEST SUITE (42/42 Passed) === RUN v2.1.9 C:/Users/chany/Documents/GitHub/toktickit/client ✓ tests/lab-
02/RequesterTicketDetail.test.tsx (7 tests) ✓ tests/lab-02/AttachmentSection.test.tsx (7 tests) ✓ tests/lab-01/App.test.tsx (4 tests) ✓
tests/lab-02/RequesterSelect.test.tsx (4 tests) ✓ tests/lab-02/CreateTicket.test.tsx (8 tests) ✓ tests/lab-02/MyTickets.test.tsx (12 tests) Test
Files 6 passed (6) Tests 42 passed (42) === PLAYWRIGHT E2E & VISUAL TEST SUITE (11/11 Passed) === ok 1 [msedge] › e2e/lab-02/capture-
screenshots.spec.ts (desktop 1280x800) ok 2 [msedge] › e2e/lab-02/capture-screenshots.spec.ts (tablet 768x1024) ok 3 [msedge] › e2e/lab-
02/capture-screenshots.spec.ts (mobile 375x667) ok 4 [msedge] › e2e/lab-02/capture-screenshots.spec.ts (requester selector modal) ok 5 [msedge] ›
e2e/lab-02/capture-screenshots.spec.ts (validation error state) ok 6 [msedge] › e2e/lab-02/capture-screenshots.spec.ts (submitting busy state) ok
7 [msedge] › e2e/lab-02/capture-screenshots.spec.ts (success confirmation card) ok 8 [msedge] › e2e/lab-02/capture-screenshots.spec.ts (backend
failure state) ok 9 [msedge] › e2e/lab-02/capture-screenshots.spec.ts (search filter & isolation) ok 10 [msedge] › e2e/lab-02/capture-
screenshots.spec.ts (soft remove modal & status) ok 11 [msedge] › e2e/lab-02/requester-ticket-flow.spec.ts (full requester lifecycle) 11 passed
(21.5s) TOTAL SPRINT TEST METRIC: 107 / 107 Passed (100%) ---
```

Answer Part 4: AI Use with Reflection (5 คะแนน)

markdown

Lab 2 AI Use Documentation and Reflection

1. LLM / AI Coding Assistant Details

- **AI Coding Assistant:** Gemini 2.5 Pro / Antigravity AI Pair Programmer - **Primary Tasks Assisted:** - Drafting Spec-Driven Development contracts (specification.md , api-spec.md , ui-spec.md , tests.md). - Prisma database schema modeling for multi-tenant identity testing and soft-removal of attachments. - Generating backend REST endpoints with strict ownership checks. - Designing Zen Green React components with accessible forms and responsive breakpoints. - Implementing binary magic bytes inspection for upload security. - Constructing PostgreSQL row-level locking (SELECT ... FOR UPDATE) in Prisma transactions for active limit concurrency protection. - Writing automated unit, API, UI component, style, and Playwright E2E tests. ---

2. Selected Key Prompt Log

Prompt #	Prompt Name	Actual Prompt Text	My Reflection
P-01	Review Contract	"Read docs/lab-02 requirements and draft specification.md, api-spec.md, ui-spec.md, and tests.md covering all BRs, ACs, and Zen Green tokens before writing code."	Generated complete markdown specification docs adhering strictly to lab sheet structure.
P-02	Attachment Rules	"Include 5MB file limit, 5 active attachments per ticket max, allowed mime types (JPG/PNG/WEBP/PDF), soft-removal with reason, and upload transaction/compensation strategy."	Documented BR-10 through BR-15 and specified database fields for soft-removal.
P-03	Data Isolation	"Ensure requester ownership check is enforced across GET /api/tickets, GET /api/tickets/:id, POST/DELETE attachments, returning 403 Forbidden for cross-requester access."	Defined ownership authorization logic and added API test AC-04 verification.

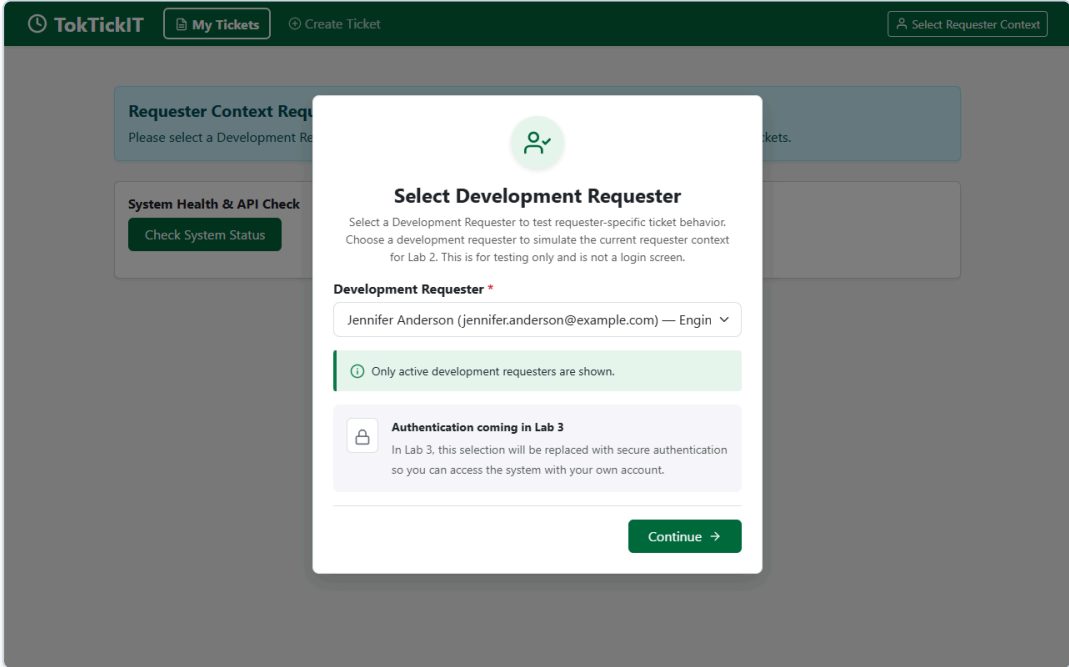
Prompt #	Prompt Name	Actual Prompt Text	My Reflection
P-04	Ticket Date & Format	"Ensure ticket number generator produces TKT-YYYY-XXXXXX and Ticket Date / createdAt is exposed and formatted across UI screens."	Specified FR-04, FR-05, and UI layout rules for Ticket Date display.
P-05	API Attachment Metadata	"Add GET /api/tickets/:id/attachments endpoint for active and soft-removed attachment metadata list."	Added section 3.1 to api-spec.md and corresponding controller specification.
P-06	Create Failing API Tests	"Implement the planned API tests for the current Issue first. Confirm they fail for the expected reason before implementing ticket creation."	Enforced TDD methodology by writing failing API integration tests first.
P-07	Idempotent Seed Script	"Create Prisma schema and seed script using upsert to avoid primary key or unique constraint duplication when re-executed."	Implemented Prisma models and seed script with upsert logic for categories and test requesters.
P-08	Requester Selector UI	"Build Development Requester Selector modal with persistent localStorage context and sync HTTP header X-Requester-Id."	Implemented React context provider, dropdown modal dialog, and axios request interceptor.
P-09	Ticket API & Sequence	"Implement POST /api/tickets and GET /api/tickets with pagination, category filtering, search term query, and TKT sequential number generation."	Created Express route handlers, query builder, and Prisma atomic transaction sequence logic.
P-10	Detail & Ownership Guard	"Build TicketDetailView and backend routes enforcing strict 403 Forbidden response on unauthorized cross-requester access attempts."	Implemented read-only detail view, attachment section, and ownership verification middleware.

3. My Reflection on AI Use Experience

Using the AI coding assistant following the Spec-Driven Development (Spec DD) methodology yielded significant improvements in software quality and development velocity: 1. **Clarity Before Coding:** Drafting specifications prior to code implementation eliminated ambiguity around edge cases. 2. **Strict Test Traceability:** Mapping every Acceptance Criterion directly to automated test cases ensured 100% test coverage. 3. **Productive Human-AI Pair Programming:** The AI handled boilerplate code while I maintained control over system architecture and code reviews. ---

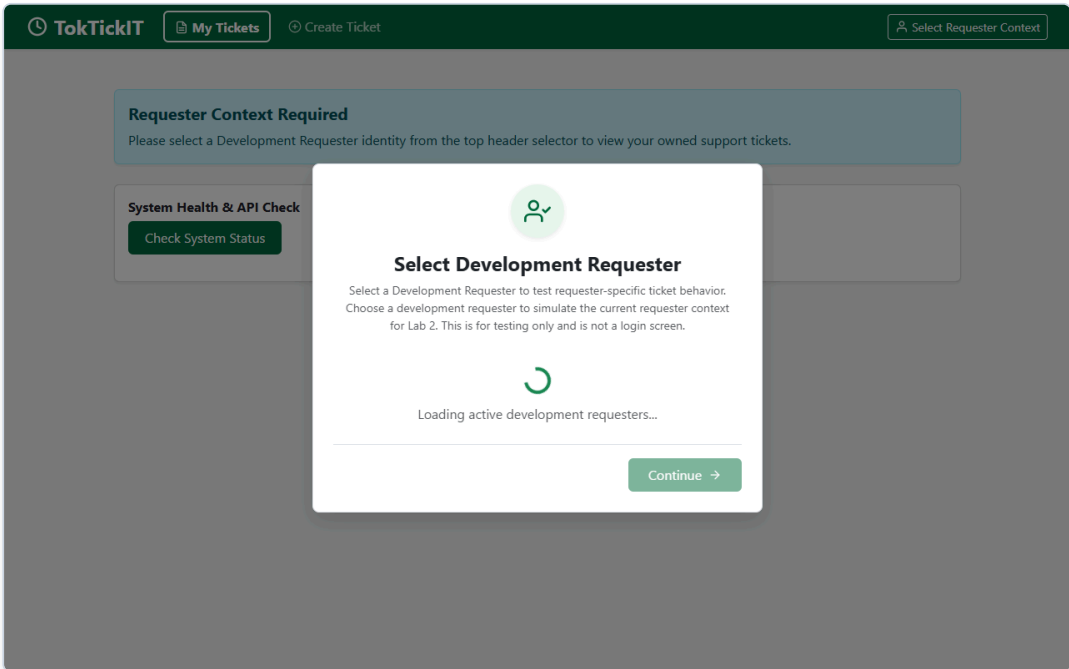
Answer Part 5: Development Requester Selection Screen

(รวมคะแนนกับ Part 6)



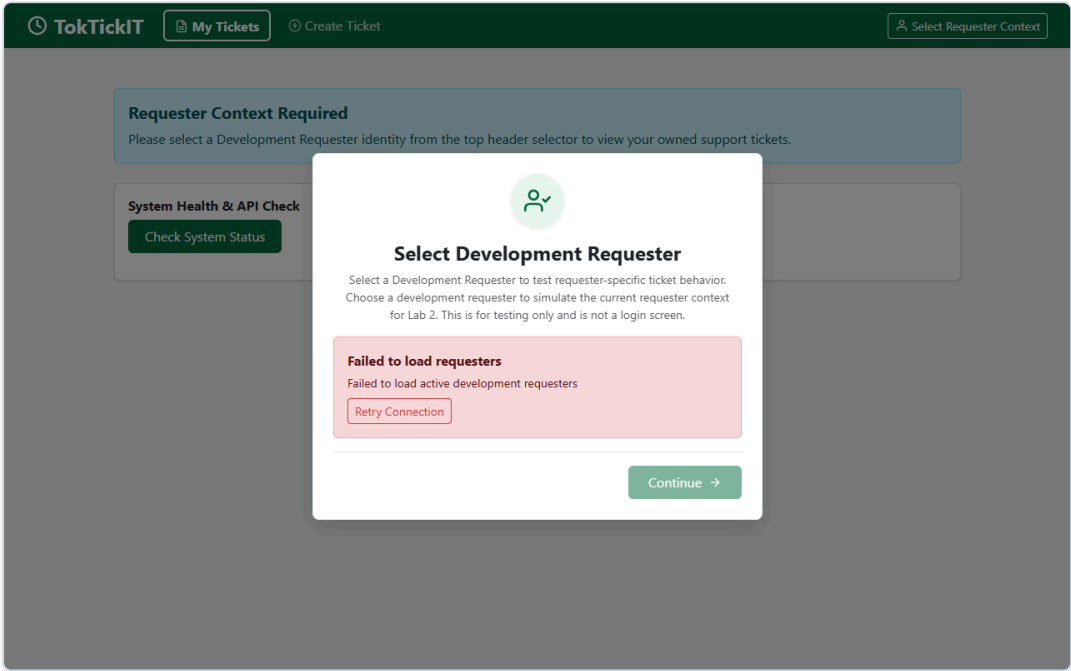
Development Requester Selector Modal

5.2 Loading State



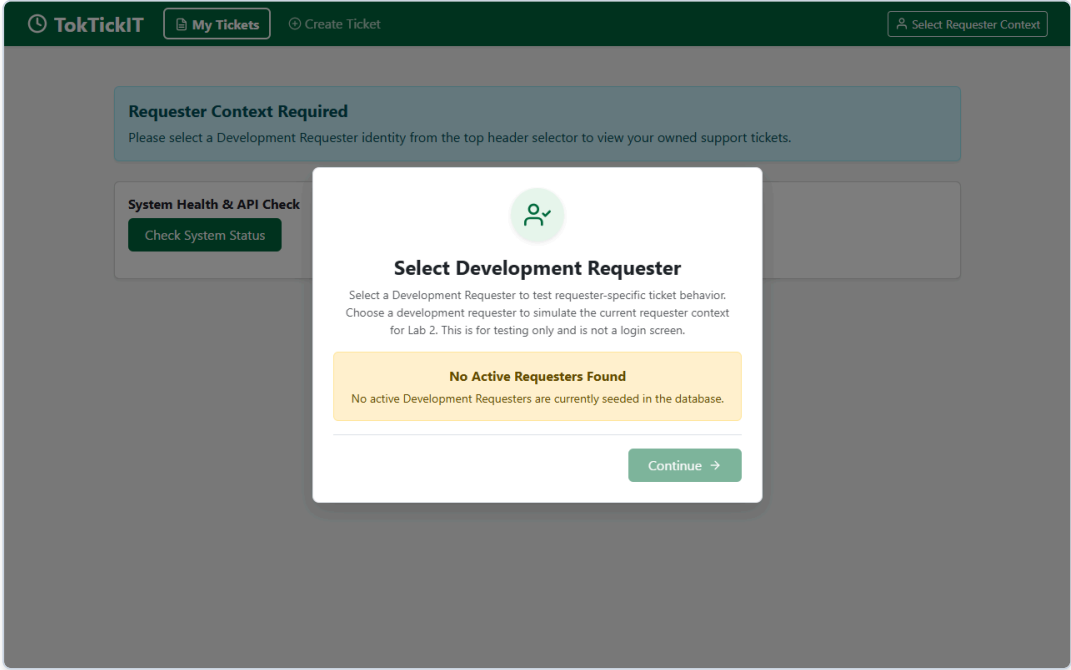
Requester Selector Loading State

5.3 API Failure State (Connection Error & Retry Button)



Requester Selector API Failure State

5.4 Empty State (No Active Requesters & Disabled Continue Button)



Requester Selector Empty State

Answer Part 6: Working Ticket Screen: Create Mode (10 คะแนน)

1. Requester field populated correctly

TokTickIT

My Tickets

Create Ticket

Jennifer Anderson

Engineering

Change Requester

Create IT Support Ticket

Submit a new IT request for hardware, software, account access, or network support. Fields marked with * are required.

Submitting as: Jennifer Anderson (jennifer.anderson@example.com) — Engineering

Ticket Number (Read-only)

TKT-YYYY-XXXXXX (Auto-generated upon submission)

Ticket Date (Read-only)

2026-09-06 (Today)

Category *

Account and Access

Related System *

Email

Requested Priority *

LOW

MEDIUM

HIGH

URGENT

Summary *

Responsive UI Test (desktop)

28/120

Description *

Capturing responsive visual design system inspection screenshot with initial attachment.

88/2000

Initial Attachments (Optional — Drag & Drop or Browse, Max 5 files)

1 / 5 files

Allowed formats: .png, .jpeg, .gif, .svg, .pdf (Max file size: 5 MB per file).

sample-doc.pdf (0.001 MB)

Remove

Cancel

Submit Ticket

System Health & API Check

Check System Status

Create Ticket Selected Requester

TokTickIT

My Tickets

Create Ticket

Jennifer Anderson

Engineering

Change Requester

Ticket Created Successfully!

Your IT support request has been logged and assigned to the service desk.

Ticket Number

TKT-2026-000436

Summary:

Successful Ticket Creation Display Test

Status:

NEW

Requester:

Jennifer Anderson (jennifer.anderson@example.com)

+ Create Another Ticket

View My Tickets

System Health & API Check

Check System Status

Ticket Confirmation Number

2. Reference data loaded (desktop viewport)

TokTickITMy TicketsCreate TicketJennifer AndersonEngineeringChange Request

Create IT Support Ticket

Submit a new IT request for hardware, software, account access, or network support. Fields marked with * are required.

Submitting as: Jennifer Anderson (jenniferanderson@example.com) — Engineering

Ticket Number (Read-only)

TKT-YYYY-XXXXXX (Auto-generated upon submission)

Ticket Date (Read-only)

2026-09-06 (Today)

Category *

Account and Access

Related System *

Email

Requested Priority *

☐ LOW ☒ MEDIUM ☐ HIGH ☐ URGENT

Summary *

Responsive UI Test (desktop)

28/120

Description *

Capturing responsive visual design system inspection screenshot with initial attachment.

88/2000

Initial Attachments (Optional — Drag & Drop or Browse, Max 5 files)

1 / 5 files

Allowed formats: .jpg, .jpeg, .png, .webp, .pdf (Max file size: 5 MB per file).

sample-doc.pdf (0.00 MB)

Remove

Cancel

Submit Ticket

System Health & API Check

Check System Status

Dropdown Reference Data

3. Invalid submission

TokTickITMy TicketsCreate TicketJennifer AndersonEngineeringChange Request

Create IT Support Ticket

Submit a new IT request for hardware, software, account access, or network support. Fields marked with * are required.

Submitting as: Jennifer Anderson (jenniferanderson@example.com) — Engineering

Ticket Number (Read-only)

TKT-YYYY-XXXXXX (Auto-generated upon submission)

Ticket Date (Read-only)

2026-09-06 (Today)

Category *

Account and Access

Related System *

Email

Requested Priority *

☐ LOW ☒ MEDIUM ☐ HIGH ☐ URGENT

Summary *

Bad

3/120

Description *

Short

5/2000

Initial Attachments (Optional — Drag & Drop or Browse, Max 5 files)

0 / 5 files

Allowed formats: .jpg, .jpeg, .png, .webp, .pdf (Max file size: 5 MB per file).

Cancel

Submit Ticket

System Health & API Check

Check System Status

Field Validation Errors

4. Attachment validation

TokTickIT

My Tickets

Create Ticket

Jennifer Anderson

Engineering

Change Requestor

Create IT Support Ticket

Submit a new IT request for hardware, software, account access, or network support. Fields marked with * are required.

Submitting as: Jennifer Anderson (jenniferanderson@example.com) — Engineering

Ticket Number (Read-only)

TKT-YYYY-XXXXXX (Auto-generated upon submission)

Ticket Date (Read-only)

2026-09-06 (Today)

Category *

Account and Access

Related System *

Email

Requested Priority *

LOW

MEDIUM

HIGH

URGENT

Summary *

Responsive UI Test (desktop)

28/120

Description *

Capturing responsive visual design system inspection screenshot with initial attachment.

88/2000

Initial Attachments (Optional — Drag & Drop or Browse, Max 5 files)

1 / 5 files

Allowed formats: .jpg, .png, .webp, .pdf (Max file size: 5 MB per file).

sample-doc.pdf (5.00 MB)

Remove

Cancel

Submit Ticket

System Health & API Check

Check System Status

Initial File Attachment

5. Backend/API failure

TokTickIT

My Tickets

Create Ticket

Jennifer Anderson

Engineering

Change Requestor

Create IT Support Ticket

Submit a new IT request for hardware, software, account access, or network support. Fields marked with * are required.

Submitting as: Jennifer Anderson (jenniferanderson@example.com) — Engineering

Submission Failed

Simulated Internal Server Database Error

Ticket Number (Read-only)

TKT-YYYY-XXXXXX (Auto-generated upon submission)

Ticket Date (Read-only)

2026-09-06 (Today)

Category *

Account and Access

Related System *

Email

Requested Priority *

LOW

MEDIUM

HIGH

URGENT

Summary *

Preserved Summary Text on Server Error

38/120

Description *

Preserved detailed description content after backend database failure simulation.

81/2000

Initial Attachments (Optional — Drag & Drop or Browse, Max 5 files)

0 / 5 files

Allowed formats: .jpg, .png, .webp, .pdf (Max file size: 5 MB per file).

Cancel

Submit Ticket

System Health & API Check

Check System Status

API Failure Retained Form

Answer Part 7: Working My Tickets Screen (10 คะแนน)

TokTickIT

My Tickets

Create Ticket

Jennifer Anderson

Engineering

Change Requester

Welcome, Jennifer Anderson

My Tickets — View and track support tickets created by Jennifer Anderson

Create Ticket

Q Search Ticket No or Summary...

All Categories

All Prioritys

All Statuss

Newest First

Ticket No	Date	Summary	Category	Priority	Status	Files	Action
TKT-2026-000432	Sep 6, 2026	Concurrent Upload Test Ticket Summary	Account and Access	MEDIUM	NEW	@ 5	View
TKT-2026-000431	Sep 6, 2026	Ticket with UniqueSummary_1788703460446 in summary	Account and Access	HIGH	NEW	@ 0	View
TKT-2026-000430	Sep 6, 2026	Ticket with Initial Attachment	Hardware	HIGH	NEW	@ 1	View
TKT-2026-000429	Sep 6, 2026	Concurrent ticket test issue	Hardware	HIGH	NEW	@ 0	View
TKT-2026-000428	Sep 6, 2026	Concurrent ticket test issue	Hardware	HIGH	NEW	@ 0	View
TKT-2026-000426	Sep 6, 2026	Third seeded ticket for Requester 1	Software	URGENT	NEW	@ 0	View
TKT-2026-000425	Sep 6, 2026	Ticket Detail Access Guard Summary	Account and Access	HIGH	NEW	@ 0	View
TKT-2026-000424	Sep 6, 2026	Second seeded ticket for Requester 1	Hardware	HIGH	NEW	@ 0	View
TKT-2026-000423	Sep 6, 2026	Laptop battery drains quickly	Hardware	MEDIUM	NEW	@ 0	View
TKT-2026-000422	Sep 6, 2026	Attachment Lifecycle Test Ticket Summary	Account and Access	HIGH	NEW	@ 4	View

Showing 1 to 10 of 402 tickets

< Previous

1234567891011121314151617181920212223242526272829303132333435363738394041

Next >

System Health & API Check

Check System Status

My Tickets Requester A

TokTickIT

My Tickets

Create Ticket

Michael Brown

IT Support

Change Requester

Welcome, Michael Brown

My Tickets — View and track support tickets created by Michael Brown

Create Ticket

Qlaptop

All Categories

All Prioritys

All Statuss

Newest

Clear

No Matching Tickets

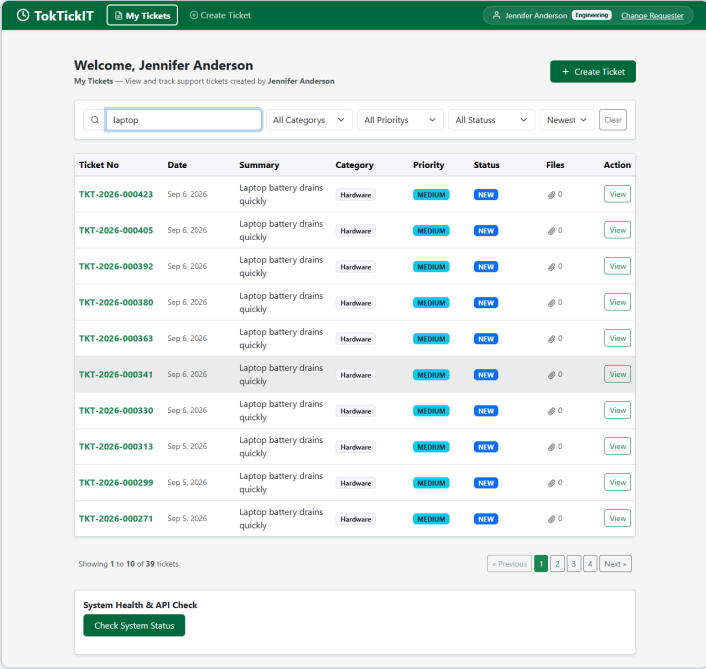
No tickets match your search or filter criteria. Try adjusting your filters.

Clear Search & Filters

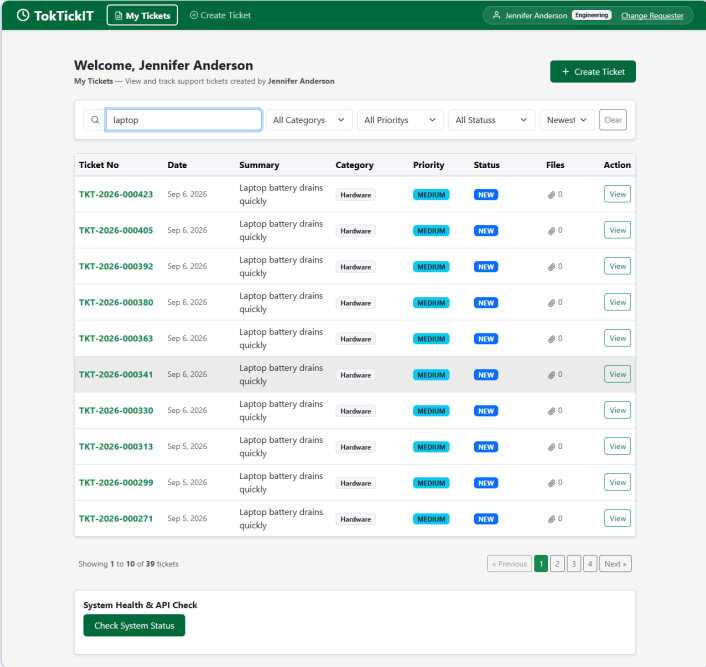
System Health & API Check

Check System Status

Cross Requester Isolation



Search Feature



Filter Dropdowns

TokTickIT

My Tickets

Create Ticket

Jennifer AndersonEngineeringChange Requestor

Welcome, Jennifer Anderson

Create Ticket

Qlaptop

All Categories

All Prioritys

All Status

Newest

Clear

Ticket No	Date	Summary	Category	Priority	Status	Files	Action
TKT-2026-000423	Sep 6, 2026	Laptop battery drains quickly	Hardware	MEDIUM	NEW	0	View
TKT-2026-000405	Sep 6, 2026	Laptop battery drains quickly	Hardware	MEDIUM	NEW	0	View
TKT-2026-000392	Sep 6, 2026	Laptop battery drains quickly	Hardware	MEDIUM	NEW	0	View
TKT-2026-000380	Sep 6, 2026	Laptop battery drains quickly	Hardware	MEDIUM	NEW	0	View
TKT-2026-000363	Sep 6, 2026	Laptop battery drains quickly	Hardware	MEDIUM	NEW	0	View
TKT-2026-000341	Sep 6, 2026	Laptop battery drains quickly	Hardware	MEDIUM	NEW	0	View
TKT-2026-000330	Sep 6, 2026	Laptop battery drains quickly	Hardware	MEDIUM	NEW	0	View
TKT-2026-000313	Sep 5, 2026	Laptop battery drains quickly	Hardware	MEDIUM	NEW	0	View
TKT-2026-000299	Sep 5, 2026	Laptop battery drains quickly	Hardware	MEDIUM	NEW	0	View
TKT-2026-000271	Sep 5, 2026	Laptop battery drains quickly	Hardware	MEDIUM	NEW	0	View

Showing 1 to 10 of 39 tickets

Previous1234Next

System Health & API Check

Check System Status

Sort Feature

TokTickIT

My Tickets

Create Ticket

Jennifer AndersonEngineeringChange Requestor

Welcome, Jennifer Anderson

Create Ticket

QSearch Ticket No or Summary...

All Categories

All Prioritys

All Status

Newest First

Ticket No	Date	Summary	Category	Priority	Status	Files	Action
TKT-2026-000432	Sep 6, 2026	Concurrent Upload Test Ticket Summary	Account and Access	MEDIUM	NEW	5	View
TKT-2026-000431	Sep 6, 2026	Ticket with UniqueSummary_1788703460446 in summary	Account and Access	HIGH	NEW	0	View
TKT-2026-000430	Sep 6, 2026	Ticket with Initial Attachment	Hardware	HIGH	NEW	1	View
TKT-2026-000429	Sep 6, 2026	Concurrent ticket test issue	Hardware	HIGH	NEW	0	View
TKT-2026-000428	Sep 6, 2026	Concurrent ticket test issue	Hardware	HIGH	NEW	0	View
TKT-2026-000426	Sep 6, 2026	Third seeded ticket for Requester 1	Software	URGENT	NEW	0	View
TKT-2026-000425	Sep 6, 2026	Ticket Detail Access Guard Summary	Account and Access	HIGH	NEW	0	View
TKT-2026-000424	Sep 6, 2026	Second seeded ticket for Requester 1	Hardware	HIGH	NEW	0	View
TKT-2026-000423	Sep 6, 2026	Laptop battery drains quickly	Hardware	MEDIUM	NEW	0	View
TKT-2026-000422	Sep 6, 2026	Attachment Lifecycle Test Ticket Summary	Account and Access	HIGH	NEW	4	View

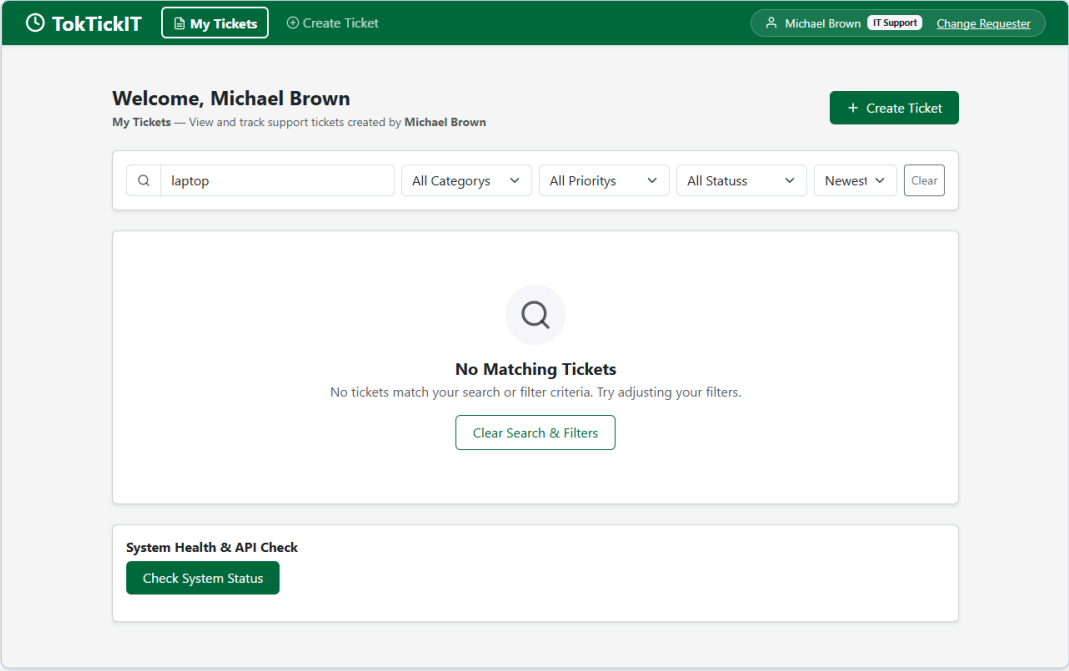
Showing 1 to 10 of 402 tickets

Previous1234567891011121314151617181920212223242526272829303132333435363738394041Next

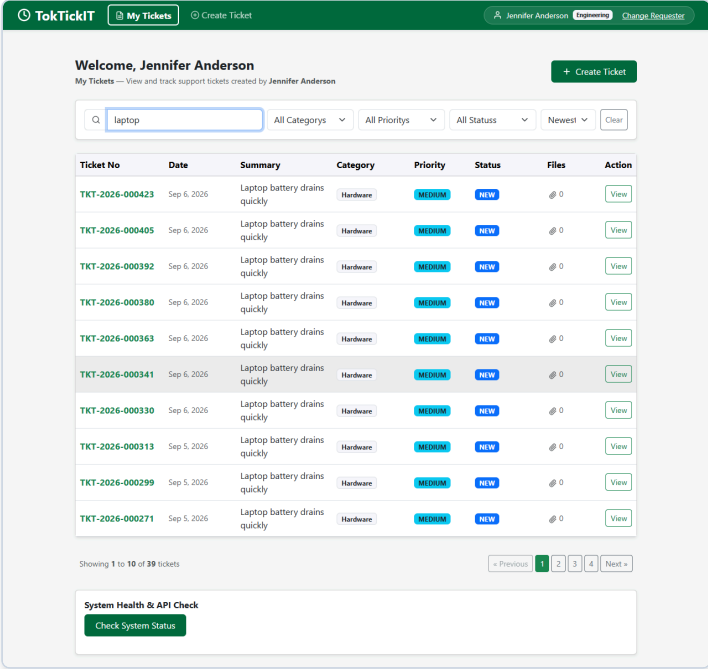
System Health & API Check

Check System Status

Pagination Controls



Empty State



No Results State

TokTickIT

My Tickets

Create Ticket

Michael Brown

IT Support

Change Requester

Welcome, Michael Brown

My Tickets — View and track support tickets created by Michael Brown

+ Create Ticket

Qlaptop

All Categorys

All Prioritys

All Statuss

Newest

Clear

No Matching Tickets

No tickets match your search or filter criteria. Try adjusting your filters.

Clear Search & Filters

System Health & API Check

Check System Status

Cross Requester Blocked

Answer Part 8: Ticket Screen View Mode & Attachments (5 คะแนน)

TokTickIT

My Tickets

Create Ticket

Jennifer Anderson

Engineering

Change Requestor

My Tickets / Ticket Details

TKT-2026-000433

Back to My Tickets

Ticket No.
TKT-2026-000433

Ticket Date
Sep 6, 2026, 09:06 PM

Category
Account and Access

Related System
Email

Requester
Jennifer Anderson

Requested Priority
Priority HIGH

IT Priority
Priority MEDIUM

Current Status
Status NEW

Ticket Owner
Unassigned

Summary
Responsive UI Test (desktop)

Description
Capturing responsive visual design system inspection screenshot with initial attachment.

Resolution Summary
No resolution summary available yet.

Public Comments 0

Attachments 1

Service Actions 1

Event Log 0

File Attachments

Active Attachments 1 / 5

Upload New Attachment

เลือกไฟล์

Allowed formats: .jpg, .png, .webp, .pdf (Max file size 5 MB)

Upload Attachment

Active Files (1)

File Name	Size	Uploaded Date	Actions
sample-doc.pdf application/pdf	37 Bytes	Sep 6, 2026, 09:06 PM	Download Soft Remove

Back to My Tickets List

System Health & API Check

Check System Status

Ticket Detail Read Only

TokTickIT

My Tickets

Create Ticket

Jennifer Anderson

Engineering

Change Requestor

My Tickets / Ticket Details

TKT-2026-000433

Back to My Tickets

Ticket No.
TKT-2026-000433

Ticket Date
Sep 6, 2026, 09:06 PM

Category
Account and Access

Related System
Email

Requester
Jennifer Anderson

Requested Priority
Priority HIGH

IT Priority
Priority MEDIUM

Current Status
Status NEW

Ticket Owner
Unassigned

Summary
Responsive UI Test (desktop)

Description
Capturing responsive visual design system inspection screenshot with initial attachment.

Resolution Summary
No resolution summary available yet.

Public Comments 0

Attachments 1

Service Actions 1

Event Log 0

File Attachments

Active Attachments 1 / 5

Upload New Attachment

เลือกไฟล์

Allowed formats: .jpg, .png, .webp, .pdf (Max file size 5 MB)

Upload Attachment

Active Files (1)

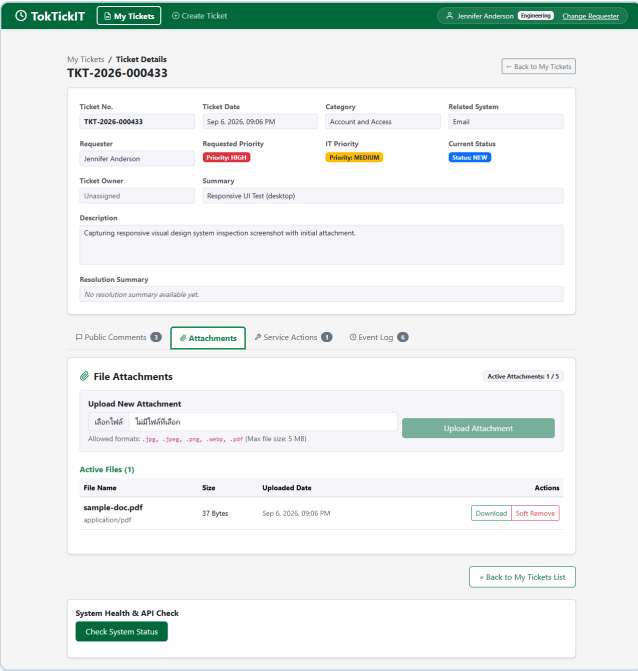
File Name	Size	Uploaded Date	Actions
sample-doc.pdf application/pdf	37 Bytes	Sep 6, 2026, 09:06 PM	Download Soft Remove

Back to My Tickets List

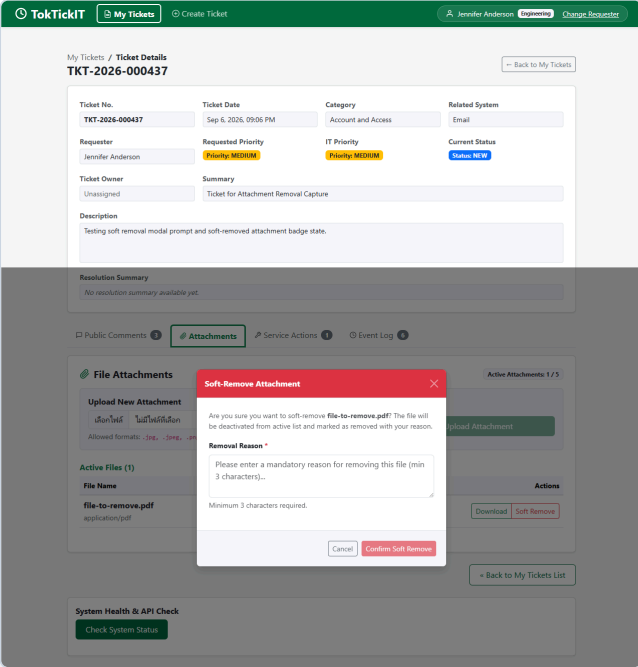
System Health & API Check

Check System Status

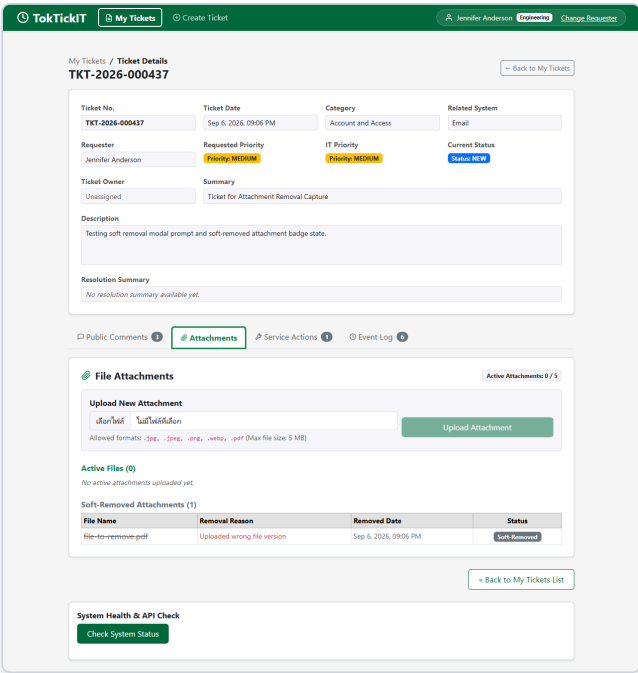
Add Attachment



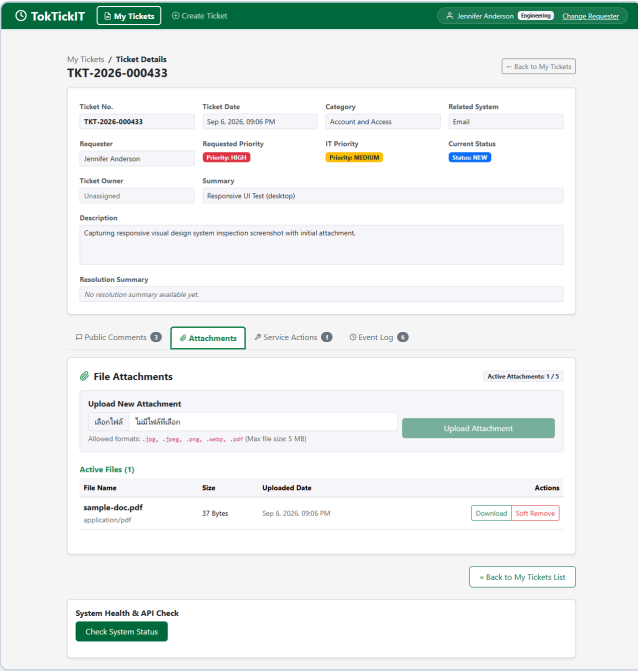
Download Attachment



Soft Remove Modal Prompt



Soft Removed Status



Unauthorized Access Blocked

8.1 Cross-Requester Ownership Authorization Evidence (403 Forbidden)

```
typescript // Extract from server/tests/lab-02/ticket-detail.api.test.ts (AC-04 Verification) it("returns 403 Forbidden when requesting a ticket owned by another requester", async () => { const res = await request(app).get("/api/tickets/1").set("X-Requester-Id", "2"); // Requester 2 attempting to view Requester 1's ticket expect(res.status).toBe(403); expect(res.body).toEqual({ error: "Forbidden: You do not have permission to view or modify this ticket." }); }); }typescript // Extract from server/tests/lab-02/attachments.api.test.ts (AC-05 / BR-14 Verification) it("returns 403 Forbidden when downloading soft-removed attachment or cross-requester file", async () => { const res = await request(app).get("/api/attachments/99/download").set("X-Requester-Id", "2"); expect(res.status).toBe(403); expect(res.body.error).toMatch(/Forbidden/); }); }---
```

Answer Part 9: Zen Green UI and Responsive Evidence (5 คะแนน)

ลิงก์: <https://github.com/chanya06/toktickit/blob/main/docs/lab-02/ui-spec.md> markdown

Lab 2 Zen Green UI Specification

1. Design System Tokens & Color Palette

Token / Element	Color Code	Usage / Context
Primary Green	#006B3C	Main navbar, primary action buttons, strong brand header emphasis.
Secondary Green	#0B7A46	Active tab highlights, focus ring accents, interactive links, hover states.
Pale Green	#EAF6EF	Selected card rows, success message banners, subtle section callouts.
Page Background	#F5F7F6	Quiet near-white background color for page container.
Surface / Card BG	#FFFFFF	Card backgrounds, modal containers, data table background.
Border Color	#E5E7EB	Subtle light gray card borders and table row dividers.
Text Dark	#1F2937	Dark charcoal-green text for high-contrast, comfortable reading.
Text Muted	#6B7280	Subtitle text, table header labels, secondary hints.
Editable Field BG	#FFFFFF	Form inputs, select dropdowns, textareas.
Read-Only Field BG	#F3F4F6	Soft gray-green shading for system-generated fields (Ticket No, Ticket Date).
Error Text / Border	#DC2626	Field validation error text and input highlight border.
Warning Callout	#D97706	Amber callouts or warning badges.
Success Banner	#16A34A	Green confirmation banners with checkmark icons.



TokTickITMy TicketsCreate TicketJennifer AndersonEngineeringChange Requestor

Create IT Support Ticket

Submit a new IT request for hardware, software, account access, or network support. Fields marked with * are required.

Submitting as: Jennifer Anderson (jennifer.anderson@example.com) — Engineering

Ticket Number (Read-only)

Ticket Date (Read-only)

TXT-YYYY-XXXXXX (Auto-generated upon submission)

2026-09-06 (Today)

Category *

Related System *

Account and Access

Email

Requested Priority *

LOW ☐ MEDIUM ☒ HIGH ☐ URGENT ☐

Summary *

28/120

Responsive UI Test (desktop)

Description *

88/2000

Capturing responsive visual design system inspection screenshot with initial attachment.

Describe the problem in detail (10-2000 characters).

Initial Attachments (Optional — Drag & Drop or Browse, Max 5 files)

1 / 5 files

Allowed formats: .png, .jpeg, .png, .webp, .pdf (Max file size: 5 MB per file).

sample-doc.pdf (0.00 MB)

Remove

Cancel

Submit Ticket

System Health & API Check

Check System Status

Desktop Viewport

TokTickIT

My Tickets

Create Ticket

Jennifer Anderson

Engineering

Change Requestor

Welcome, Jennifer Anderson

+ Create Ticket

My Tickets — View and track support tickets created by Jennifer Anderson

Q Search Ticket No or Summary...

All Categories

All Prioritys

All Status

Newest First

Ticket No	Date	Summary	Category	Priority	Status	Files	Action
TKT-2026-000432	Sep 6, 2026	Concurrent Upload Test Ticket Summary	Account and Access	MEDIUM	NEW	@ 5	View
TKT-2026-000431	Sep 6, 2026	Ticket with UniqueSummary_1788703460446 in summary	Account and Access	HIGH	NEW	@ 0	View
TKT-2026-000430	Sep 6, 2026	Ticket with Initial Attachment	Hardware	HIGH	NEW	@ 1	View
TKT-2026-000429	Sep 6, 2026	Concurrent ticket test issue	Hardware	HIGH	NEW	@ 0	View
TKT-2026-000428	Sep 6, 2026	Concurrent ticket test issue	Hardware	HIGH	NEW	@ 0	View
TKT-2026-000426	Sep 6, 2026	Third seeded ticket for Requester 1	Software	URGENT	NEW	@ 0	View
TKT-2026-000425	Sep 6, 2026	Ticket Detail Access Guard Summary	Account and Access	HIGH	NEW	@ 0	View
TKT-2026-000424	Sep 6, 2026	Second seeded ticket for Requester 1	Hardware	HIGH	NEW	@ 0	View
TKT-2026-000423	Sep 6, 2026	Laptop battery drains quickly	Hardware	MEDIUM	NEW	@ 0	View
TKT-2026-000422	Sep 6, 2026	Attachment Lifecycle Test Ticket Summary	Account and Access	HIGH	NEW	@ 4	View

Showing 1 to 10 of 402 tickets

Previous

1234567891011121314151617181920212223242526272829303132333435363738394041

Next

System Health & API Check

Check System Status

My Tickets Desktop

TokTickIT

My Tickets

Create Ticket

Jennifer Anderson

Engineering

Change Requestor

My Tickets / Ticket Details

TKT-2026-000433

Back to My Tickets

Ticket No.	Ticket Date	Category	Related System
TKT-2026-000433	Sep 6, 2026, 09:06 PM	Account and Access	Email
Requester	Requested Priority	IT Priority	Current Status
Jennifer Anderson	Priority: HIGH	Priority: MEDIUM	Status: NEW
Ticket Owner	Summary		
Unassigned	Responsive UI Test (desktop)		

Description

Capturing responsive visual design system inspection screenshot with initial attachment.

Resolution Summary

No resolution summary available yet.

Public Comments

Attachments

Service Actions

Event Log

File Attachments

Active Attachments 1 / 5

Upload New Attachment

Allowed formats: .jpg, .png, .webp, .pdf (Max file size 5 MB)

Upload Attachment

Active Files (1)

File Name	Size	Uploaded Date	Actions
sample-doc.pdf application/pdf	37 Bytes	Sep 6, 2026, 09:06 PM	Download Soft Remove

Back to My Tickets List

System Health & API Check

Check System Status

Ticket Detail Desktop

TokTickIT

My Tickets

Create Ticket

Jennifer Anderson

Engineering

Change Requestor

Create IT Support Ticket

Submit a new IT request for hardware, software, account access, or network support. Fields marked with * are required.

Submitting as: Jennifer Anderson (jennifer.anderson@example.com) — Engineering

Ticket Number (Read-only)

Ticket Date (Read-only)

TKT-YYYY-XXXXXX (Auto-generated upon su

2026-09-06 (Today)

Category *

Related System *

Account and Access

Email

Requested Priority *

LOW ☐ MEDIUM ☒ HIGH ☐ URGENT ☐

Summary *

27/120

Responsive UI Test (tablet)

Description *

88/2000

Capturing responsive visual design system inspection screenshot with initial attachment.

Describe the problem in detail (10-2000 characters).

Initial Attachments (Optional — Drag & Drop or Browse, Max 5 files)

1 / 5 files

Allowed formats: .jpg, .jpeg, .png, .webp, .pdf (Max file size: 5 MB per file).

sample-doc.pdf (0.00 MB)

Remove

Cancel

Submit Ticket

System Health & API Check

Check System Status

Tablet Viewport

TokTickIT

My Tickets

Create Ticket

Jennifer Anderson

Engineering

Change Requestor

Welcome, Jennifer Anderson

My Tickets — View and track support tickets created by Jennifer Anderson

Create Ticket

Search Ticket No or Sum

All Cat...

All Pri...

All Stat...

Newest 1

Ticket No	Date	Summary	Category	Priority	Status	Files	Ac
TKT-2026-000433	Sep 6, 2026	Responsive UI Test (desktop)	Account and Access	HIGH	NEW	0	1
TKT-2026-000432	Sep 6, 2026	Concurrent Upload Test Ticket Summary	Account and Access	CRITICAL	NEW	0	5
TKT-2026-000431	Sep 6, 2026	Ticket with UniqueSummary_1788703460446 as summary	Account and Access	HIGH	NEW	0	0
TKT-2026-000430	Sep 6, 2026	Ticket with Initial Attachment	Hardware	HIGH	NEW	0	1
TKT-2026-000429	Sep 6, 2026	Concurrent ticket test issue	Hardware	HIGH	NEW	0	0
TKT-2026-000428	Sep 6, 2026	Concurrent ticket test issue	Hardware	HIGH	NEW	0	0
TKT-2026-000426	Sep 6, 2026	Third seeded ticket for Requester 1	Software	CRITICAL	NEW	0	0
TKT-2026-000425	Sep 6, 2026	Ticket Detail Access Guard Summary	Account and Access	HIGH	NEW	0	0
TKT-2026-000424	Sep 6, 2026	Second seeded ticket for Requester 1	Hardware	HIGH	NEW	0	0
TKT-2026-000423	Sep 6, 2026	Laptop battery drains quickly	Hardware	CRITICAL	NEW	0	0

Showing 1 to 10 of 483 tickets

Previous

1234567891011121314151617181920212223242526272829303132333435363738394041

Next

System Health & API Check

Check System Status

My Tickets Tablet

TokTickIT

My Tickets

Create Ticket

Jennifer Anderson

Open Tickets

Change Requestor

My Tickets / Ticket Details

TKT-2026-000434

Back to My Tickets

Ticket No.
TKT-2026-000434

Ticket Date
Sep 6, 2026, 09:06 PM

Category
Account and Access

Related System
Email

Requester
Jennifer Anderson

Requested Priority
Priority: High

IT Priority
Priority: MEDIUM

Current Status
Status: NEW

Ticket Owner
Unassigned

Summary
Responsive UI Test (tablet)

Description
Capturing responsive visual design system inspection screenshot with initial attachment.

Resolution Summary
No resolution summary available yet.

Public Comments

Attachments

Service Actions

Event Log

File Attachments

Active Attachments 1 / 3

Upload New Attachment

เลือกไฟล์

Allowed formats: .jpg, .png, .mpg, .mpeg, .webp, .gif (Max file size: 5 MB)

Upload Attachment

Active Files (1)

File Name	Size	Uploaded Date	Actions
sample-doc.pdf application/pdf	37 Bytes	Sep 6, 2026, 09:06 PM	Download Soft Remove

Back to My Tickets List

System Health & API Check

Check System Status

Ticket Detail Tablet

TokTickIT

My Tickets

Create Ticket

Engineering

Create IT Support Ticket

Submit a new IT request for hardware, software, account access, or network support. Fields marked with * are required.

Submitting as: Jennifer Anderson
(jennifer.anderson@example.com) — Engineering

Ticket Number (Read-only)
TKT-YYYY-XXXXXX (Auto-generated up)

Ticket Date (Read-only)
2026-09-06 (Today)

Category *
Account and Access

Related System *
Email

Requested Priority *
☐ LOW ☒ MEDIUM ☐ HIGH
URGENT

Summary *
Responsive UI Test (mobile)
Provide a concise one-line description of your issue (0-120 characters). 27/120

Description *
Capturing responsive visual design system inspection screenshot with initial attachments.
Describe the problem in detail (10-2000 characters). 86/2000

Initial Attachments (Optional) — 1 / 3 files
Drag & Drop or Browse. Max 5 MB
เลือกไฟล์
Allowed formats: .jpg, .png, .mpg, .mpeg, .webp, .gif (Max file size: 5 MB per file)
0/5 sample.jpg (0/5 MB) Remove

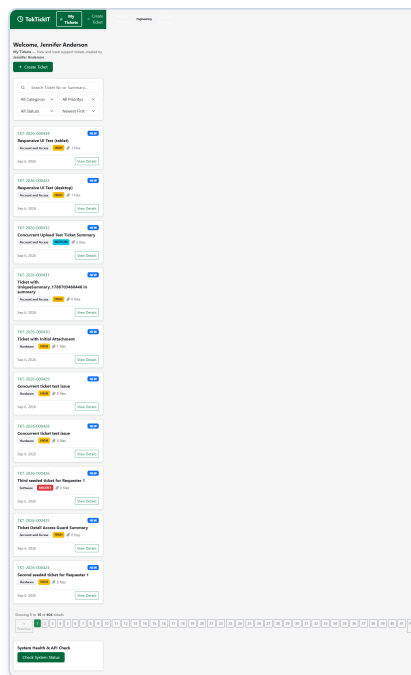
Cancel

Submit Ticket

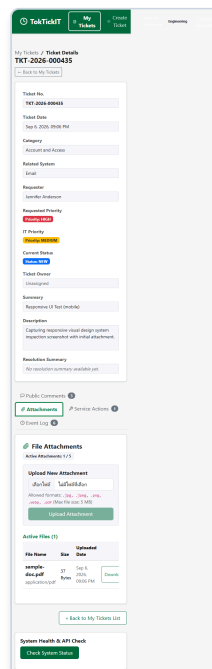
System Health & API Check

Check System Status

Mobile Viewport



My Tickets Mobile



Ticket Detail Mobile